

Public Economics

Take-Home Assignment 3

Topics: Budget Analysis, Fiscal Federalism

Name: _____

Student ID: _____

Due: _____

Answer both questions. Where a short explanation is requested, 2–3 sentences is enough — we are grading for correct reasoning, not length. Show your work for any calculation.

Question 1: Real vs. Nominal Deficits

In a given year, Austria's government runs a **nominal** budget deficit of €10 billion. Inflation that year is 3%, and the government's outstanding debt at the start of the year is €300 billion.

a) Inflation erodes the real value of existing government debt, which effectively offsets part of the nominal deficit. Calculate the **real** (inflation-adjusted) budget deficit that year. (Hint: subtract inflation $\times$ outstanding debt from the nominal deficit.)

b) A newspaper headline reads: “*Austria's deficit hits €10 billion — worse than ever!*” In 2–3 sentences, explain why reporting only the nominal deficit figure can be a misleading way to describe the government's fiscal position, drawing on your answer to part (a) and/or the idea of measuring the deficit relative to GDP.

Question 2: Tiebout and Local Public Goods

Two neighboring Austrian municipalities offer very different packages of local public services and property taxes. *Gemeinde X* has excellent local schools and high property taxes; *Gemeinde Y* has minimal local services and low property taxes.

a) Using the Tiebout “voting with your feet” logic, explain how allowing people to freely choose where to live can help reveal residents' true preferences for local public goods — something that is hard to do through voting at the national level.

b) Pick **one** realistic reason why this Tiebout sorting mechanism works less perfectly in practice than the simple theory suggests (for example: moving costs, imperfect information about what each municipality offers, zoning restrictions, or public-service spillovers that cross municipal borders). Name it and explain in 2–3 sentences why it weakens the sorting mechanism.